

Damián Pablo San Román Alerigi, PhD

Applied Physicist & R&D Lead — Photonics and High-Power Laser Systems

Upstream Advanced Research Center, Saudi Aramco

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Applied physicist with a decade of experience converting fundamental optical physics into deployed industrial technology. Co-leads Saudi Aramco's high-power laser programme, carried from first-principles modelling of laser–rock interaction through to the petroleum industry's **first industrial laser perforation and flowline-descaling field deployments**. Holds a patent portfolio of **26 granted inventions across eight jurisdictions** (11 pending), with sole or lead inventorship on ten, spanning laser downhole tools, hollow-core optical fibre, photonic/ML sensing, and hydrogen-sulfide-to-hydrogen valorisation.

26 / 11	8 + ~50	437	21	2
PATENTS GRANTED / PENDING	JOURNAL & CONFERENCE PAPERS	CITATIONS (GOOGLE SCHOLAR)	WORKS AS FIRST / LEAD AUTHOR	FIRST INDUSTRIAL DEPLOYMENTS

CORE EXPERTISE

PHYSICS & ENGINEERING

- Laser–matter interaction
- Computational electromagnetics
- Nonlinear & time-varying media
- Hollow-core fibre optics
- Photonic sensing

APPLIED / DATA

- ML for subsurface inference
- Hyperspectral sensing
- Computer vision

PROGRAMME LEADERSHIP

- Lab-to-field deployment
- IP & patent strategy
- Academic–industry collaboration

TECHNICAL TOOLKIT

LANGUAGES & TOOLS

- MATLAB
- Python
- Fortran
- C
- LaTeX
- PETSc

SOFTWARE

- COMSOL
- Mathematica
- RSoft

PROFESSIONAL EXPERIENCE

Applied Physicist — Co-PI, High-Power Laser Programme

2015 – Present · Upstream Advanced Research Center, Saudi Aramco, Dhahran, KSA

- Co-led the physics, modelling and field-execution programme that took high-power laser perforation and descaling from laboratory concept to the petroleum industry's **first industrial deployments** (2021–2024), documented across 25+ peer-reviewed and conference publications and a supporting patent portfolio.
- Physics & modelling foundation (2016–2019):** first-authored the governing numerical treatments of laser–rock interaction and the thermal/geomechanical dynamics of electromagnetic heating; the coupled thermal–mechanical model was published in *SPE Journal* (2019).
 - First-of-kind field deployments (2021–2024):** delivered the industry's first industrial flowline-descaling field trial (Dec. 2021) and first industrial high-power laser perforation deployment (2023), each with a companion field-validation study.
 - Diversification (2020–present):** opened three new research lines — hydrogen-sulfide valorisation to hydrogen, photonic/ML sensing for subsurface and surface assets, and hollow-core optical fibre for kilowatt-class downhole power delivery (with the University of Southampton Optoelectronics Research Centre).
 - Recognised by Saudi Aramco Senior Vice President Waleed A. Al-Mulhim, with the Laser Research Team, for the first laser-perforation field trial (*The Arabian Sun / Aramco Life*, 2023).

Graduate Researcher, Photonics Laboratory

2009 – 2014 · KAUST, Thuwal, KSA · Advisor: Prof. Boon S. Ooi

- PhD research on light propagation in heterogeneous, nonlinear and time-varying media — computational electromagnetics, beam shaping and chalcogenide-glass material engineering.

OriginLab

METHODS

FEM

FDTD

Spectral / paraxial methods

Inverse-problem design

EDUCATION

PhD, Electrical Engineering (Photonics)

KAUST, Saudi Arabia
2011 – 2014

M.S., Electrical Engineering (Photonics)

KAUST, Saudi Arabia —
Founding cohort
2009 – 2010

Licenciatura (B.S.), Physics

Universidad Nacional
Autónoma de México
(UNAM)
2003 – 2009 · GPA
9.67/10

IDENTIFIERS

ORCID

0000-0002-3571-6776

SCOPUS AUTHOR ID

55408065500

RESEARCHERID

N-1562-2016

LANGUAGES

Spanish (native) ·

English (fluent) · French
(working)

- Founding member of the KAUST Photonics Master's programme; developed a high-performance numerical toolbox for wave propagation in complex media (*Maxwell-GEMS*).

SELECTED INNOVATIONS (PATENTS)

US 12,682,216 B2 — Predicting rock properties using thermal sensing and machine learning **SOLE INVENTOR**

US 12,559,366 B2 — Photocatalytic conversion of hydrogen sulfide to hydrogen **LEAD INVENTOR**

US 12,140,850 B2 + EP / WO / CN / SA — Reconfigurable optics for beam transformation **LEAD · 4-JURISDICTION FAMILY**

US 11,725,504 B2 + WO / CA / EP / JP — Contactless real-time 3D mapping of surface equipment **LEAD · 5-JURISDICTION FAMILY**

US 11,555,958 B2 / EP 4,067,952 B1 — Nested anti-resonant nodeless optical fibre **WITH UNIV. OF SOUTHAMPTON**

US 12,590,836 B2 — Optical energy apparatus, systems and methods **SOLE INVENTOR**

Full record of 26 granted patents and 11 pending applications in the accompanying Extended CV.

SELECTED PUBLICATIONS

San-Roman-Alerigi *et al.*, "A possible approach on optical analogues of gravitational attractors," **Optics Express** 21(7), 8298 (2013) **FEATURED, LASER FOCUS WORLD**

San-Roman-Alerigi *et al.*, "Generation of J_0 -Bessel-Gauss beam by an heterogeneous refractive index map," **JOSA A** 29(7), 1252 (2012)

Ferree, Kosco, Laquai, Cavazos Sepúlveda, San Roman Alerigi, "Solar Fuel Production from Hydrogen Sulfide," **Adv. Energy Sustain. Res.** 4, 2200201 (2023)

Khan *et al.*, "Photochemical valorization of hydrogen sulfide," **RSC Advances** 15, 37299 (2025)

Han, Fang, San Roman Alerigi, Batarseh, "Thermal/Mechanical Interaction in Laser Perforation Process," **SPE Journal** 24(5), 2097 (2019)

Batarseh, San Roman Alerigi, Marshall, Graves, "First Industrial High-Power Laser Perforation Deployment," **SPE-215016-MS** (2023)

Full record of 12 peer-reviewed / industry journal articles and ~50 conference papers in the accompanying Extended CV.

RECOGNITION & AFFILIATIONS

- **2026** — Contributing team member on two technologies named finalists at the Gulf Energy Information Excellence Awards: *Best Pipeline Rehabilitation Technology* (High-Power Laser Descaler for Surface Flowlines) and *Best Controls, Instrumentation & Automation Technology – Upstream* (WATCH: Wellsite Asset Tracking for Control of Hazards)
- **2023** — Saudi Aramco SVP recognition (Waleed A. Al-Mulhim) for the Laser Research Team's first industrial laser-perforation field trial
- **2013** — Optics Express work featured in Laser Focus World
- **2010** — KAUST Academic Excellence Award · **2009** — KAUST Provost Award · **2008** — KAUST Discovery Scholar
- Member — IEEE · Optica (OSA) · American Physical Society · SIAM · Sociedad Mexicana de Física